

COURSEBOOK EDITION

Publication and Course Record

This edition expands the existing PHY111 coursebook while retaining its twelve-module scientific sequence. The original course already contains 24 guided exercises, 36 worked examples, twelve module quizzes, a study guide, and a formative key. Version 2.1 adds a substantial interactive layer designed around prediction, representation, derivation, experimental evidence, computation, error diagnosis, and cumulative mastery.

Course	PHY111 - Introductory Physics I: Mechanics, Fluids, Oscillations, Waves, Sound, and Gravitation
Level	Calculus-based undergraduate foundation
Prerequisites	Algebra, trigonometry, vectors, scientific notation, differential calculus, and introductory integral calculus
Recommended work-load	350-450 hours including exposition, derivations, tutorials, problem sets, laboratory or simulation work, computational activities, formal coursework, and revision
Primary scope source	Robert G. Brown, Introductory Physics I: Elementary Mechanics, Duke University Physics Department
Scientific standard	Every numerical result must be traceable to an explicit model, units, assumptions, and a check; computation is a method, not a substitute for modelling.

What Version 2.1 adds

- 24 expanded tutorial workshops
- 24 additional worked-example studios
- 12 experimental or simulation investigations
- 12 computational mini-labs
- 12 error-diagnosis tasks
- 120 additional unguided problems
- 12 additional mastery tests
- four cumulative progress examinations
- six formal assessed projects
- a new three-hour final examination
- a calculator-ready exam formula sheet
- an expanded instructor solutions and marking guide

COURSE ARCHITECTURE

From Motion to Field - Expanded Learning Cycle

The course is cumulative. Each new method is introduced as a response to a limitation in the previous description: kinematics becomes dynamics; particle dynamics becomes system conservation; translation becomes rotation; discrete bodies become continuous matter; oscillation becomes wave propagation; and local force models become a gravitational field description.

The module learning cycle

1. **Orient.** Read the module question and outcomes. Attempt the short diagnostic from memory.
2. **Build.** Study definitions and derivations. Reproduce each important derivation without looking.
3. **Represent.** Draw graphs, vectors, free-body diagrams, energy diagrams, or field sketches before algebra.
4. **Solve.** Cover worked solutions and solve first. Compare method, not only the final number.
5. **Tutorial.** Complete the expanded workshop prompts in order: prediction -> representation -> derivation -> calculation -> interpretation.
6. **Investigate.** Perform the experiment, simulation, or computational activity and compare evidence with the ideal model.
7. **Stress-test.** Debug a deliberately flawed solution and solve the less-scaffolded problem set.
8. **Assess.** Take the module mastery test closed-notes, then review errors by category.

Error categories

Model error	Wrong system, frame, approximation, omitted interaction, or unjustified conservation law.
Representation error	Graph, vector, sign convention, free-body diagram, or geometry does not match the physical situation.
Mathematical error	Algebra, calculus, trigonometry, numerical integration, or unit conversion mistake.
Interpretation error	Number is calculated correctly but physical meaning, direction, limiting case, or uncertainty is wrong.

LEARNING AND ASSESSMENT

Knowledge Testing in a Physics Course

Concept checks

Short qualitative prompts interrupt passive reading and force a prediction before calculation.

Experiments and simulations

Students must confront measurement uncertainty, non-ideal behaviour, numerical error, and the gap between model and evidence.

Worked-example cover method

Read only the problem, hide the solution, solve independently, then compare the modelling path line by line.

Error diagnosis

Deliberately flawed solutions test whether the learner can locate the first invalid inference instead of merely spotting a wrong answer.

Tutorial workshops

Structured multi-step problems require diagrams, symbolic derivation, limiting cases, and explanation.

Mastery tests

Closed-note module tests and cumulative progress exams require retrieval under time pressure.

Formal weighting

Formative work in the coursebook is ungraded. Six formal assignments contribute 60 percent of the course result. The cumulative final examination contributes 40 percent. Four progress examinations are provided for timed practice and diagnostic use but do not alter the formal weighting unless an institution adopts a different scheme.

MATHEMATICAL BOOTCAMP

Part 0 - Mathematical and Experimental Foundations

0.1 Units, dimensions, and order of magnitude

A physical equation must be dimensionally homogeneous. Unit checking is not cosmetic: it is one of the fastest ways to discover a missing factor of time, length, mass, or angle. Before evaluating a formula, write dimensions symbolically and estimate the expected order of magnitude.

Practice. Show that the dimensions of energy are ML^2T^{-2} , that power is ML^2T^{-3} , and that v^2/g has dimensions of length.

0.2 Vectors and components

A vector is independent of coordinates; its components depend on the chosen basis. Resolve vectors only after choosing axes that simplify the geometry. Dot products encode projection and work; cross products encode oriented area, torque, and angular momentum.

$$\vec{A} \cdot \vec{B} = AB \cos \theta, \quad |\vec{A} \times \vec{B}| = AB \sin \theta$$

0.3 Derivatives and integrals as physical operations

The derivative gives a local rate of change; the integral accumulates a rate over an interval. Kinematics therefore provides the first recurring calculus chain:

$$x(t) \xrightarrow{d/dt} v(t) \xrightarrow{d/dt} a(t), \quad a(t) \xrightarrow{\int dt} \Delta v \xrightarrow{\int dt} \Delta x.$$

0.4 Graphs as equations you can see

Slope and area are physical information. A graph should display axes, units, sign, scale, and uncertainty where applicable. Before using a calculator, sketch the expected graph shape from the physics.

0.5 Numerical computation

Numerical work is appropriate when an equation cannot be solved conveniently in closed form, when data are sampled, or when a model is being simulated. A computational result is incomplete without initial conditions, parameter values, units, algorithm, step size or tolerance, and a validation test against a known limit.

Minimum numerical validation. Run at least one case with a known analytic answer, repeat with a smaller time step, and verify conserved quantities when the model predicts conservation.

0.6 Measurement and uncertainty

Measurements are estimates, not exact mathematical objects. Distinguish instrument resolution, random scatter, systematic calibration error, and model error. Report enough significant figures to reflect the evidence rather than the calculator display.

Practice. A length is 1.245 ± 0.005 m and a time is 0.82 ± 0.01 s. Estimate speed and its fractional uncertainty using a first-order propagation rule.

PART I

The Full Course

Twelve modules from Newtonian model construction through gravitation. The original course exposition is retained as the core script. Each module is now followed by an expanded tutorial studio before its module quiz.

Rule for every problem: define the system -> state the assumptions -> choose the representation -> write the governing law -> solve symbolically -> evaluate -> check units, sign, limiting cases, and physical scale.

MODULE 01 • FULL COURSE

Module 1 — Newtonian Description of Motion and Force

Coordinates, vectors, free-body diagrams, and Newton's laws

How does physics turn motion into a quantitative model?

LEARNING OUTCOMES

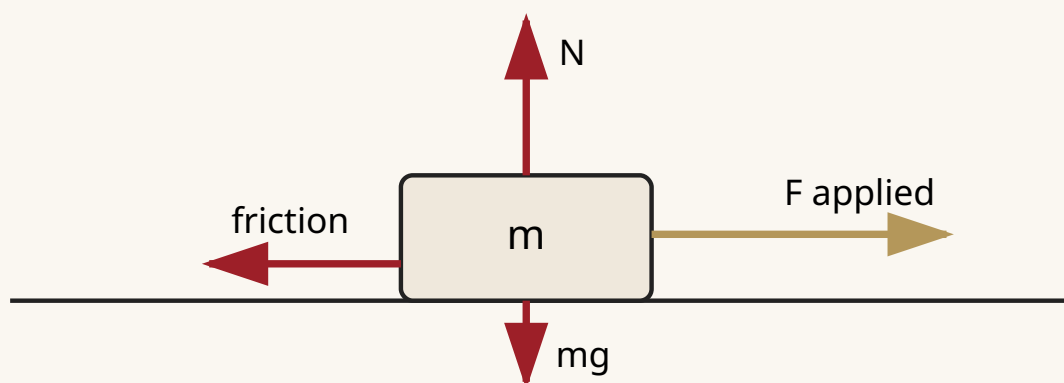
1. distinguish position, displacement, distance, velocity, and acceleration

2. resolve vectors into components and choose a useful coordinate system

3. construct a free-body diagram without adding fictitious or duplicate forces

4. apply Newton's three laws to one- and two-dimensional motion

5. translate a verbal situation into equations, solve them, and test the result



A free-body diagram isolates one object and displays only the external forces acting on it.

MODULE 01 • CONCEPTUAL DEVELOPMENT

1.1 Models, coordinates, and the language of motion

Physics begins by replacing an overwhelmingly detailed event with a deliberately limited model. A moving car can be treated as a particle when its size and rotation do not matter; Earth can be treated as a uniform gravitational field when the height change is small compared with Earth's radius. A good model is not a photograph of reality. It is a controlled abstraction that preserves the features needed to answer a question.

A coordinate system gives numerical meaning to location. In one dimension, a signed coordinate x distinguishes positions on opposite sides of an origin. In two or three dimensions, a position vector identifies a point relative to the chosen origin. The origin and positive directions are choices, not properties of nature, so changing them may change the signs of components without changing the physical event.

Displacement is the change in position and therefore depends only on the endpoints. Distance is the length of the path and is never negative. Average velocity is displacement divided by elapsed time; average speed is distance divided by elapsed time. This distinction is already a test of disciplined modeling: a runner who returns to the start has traveled a nonzero distance but has zero displacement.

$$\Delta x = x_f - x_i$$

Displacement is a signed change of coordinate.

$$\bar{v} = \Delta x / \Delta t$$

Average velocity; the bar indicates an average over a finite interval.

$$\bar{a} = \Delta v / \Delta t$$

Average acceleration measures the change of velocity, not merely a change of speed.

CONCEPT CHECK

- Why can a coordinate be negative while a distance cannot?
- Which quantities change if the positive axis is reversed?

WORKED EXAMPLE**A round trip with unequal speeds**

A student walks 120 m east in 80 s and then 120 m west in 120 s. Find average velocity and average speed.

1. Choose east as positive. The total displacement is $+120 \text{ m} - 120 \text{ m} = 0$.
2. The elapsed time is 200 s, so the average velocity is $0/200 = 0 \text{ m s}^{-1}$.
3. The total distance is 240 m, so average speed is $240/200 = 1.20 \text{ m s}^{-1}$.

Result. Average velocity = 0; average speed = 1.20 m s^{-1} .

Interpretation. The same journey can have zero average velocity and a positive average speed because velocity records net change, while speed records accumulated path length.

MODULE 01 • CONCEPTUAL DEVELOPMENT

1.2 Instantaneous velocity, acceleration, and constant-acceleration kinematics

The instantaneous velocity is the limiting slope of the position-time graph as the time interval shrinks. Acceleration is the limiting slope of the velocity-time graph. These graphical meanings are more fundamental than any memorized kinematic formula: the slope of $x(t)$ is $v(t)$, and the area under $v(t)$ is displacement.

For constant acceleration, velocity changes linearly with time and position changes quadratically. The familiar kinematic equations are not separate laws of nature; they are consequences of the definitions of velocity and acceleration under the special assumption that acceleration is constant. They must not be used uncritically when acceleration depends on time, position, or velocity.

Signs carry physical meaning. A negative velocity means motion in the negative coordinate direction. A negative acceleration means the velocity is changing toward the negative direction; it does not automatically mean that an object is slowing down. An object speeds up when velocity and acceleration have the same sign and slows down when their signs differ.

$$v_f = v_i + at$$

Valid only when acceleration is constant.

$$\Delta x = v_i t + \frac{1}{2}at^2$$

Position change under constant acceleration.

$$v_f^2 = v_i^2 + 2a\Delta x$$

Useful when time is absent from the known data.

CONCEPT CHECK

- Can velocity be zero while acceleration is nonzero? Give an example.
- What feature of a velocity-time graph gives displacement?

WORKED EXAMPLE**Braking to rest**

A bicycle travels at 8.0 m s^{-1} and brakes with constant acceleration -2.0 m s^{-2} . Find the stopping time and distance.

1. Set $v_f = 0$. From $v_f = v_i + at$, $t = (0 - 8.0)/(-2.0) = 4.0 \text{ s}$.
2. Use $\Delta x = v_i t + \frac{1}{2}at^2 = 8.0(4.0) + \frac{1}{2}(-2.0)(4.0)^2$.
3. The displacement is $32 - 16 = 16 \text{ m}$.

Result. Stopping time = 4.0 s; stopping distance = 16 m.

Interpretation. Doubling the initial speed while keeping the same braking acceleration would quadruple the braking distance.

MODULE 01 • CONCEPTUAL DEVELOPMENT

1.3 Vectors, components, and projectile motion

A vector has magnitude and direction. It can be represented geometrically by an arrow or algebraically by components. Components are not independent physical objects; they are the projections of one vector onto chosen axes. Because perpendicular components can be treated separately and recombined, vector decomposition converts many two-dimensional problems into paired one-dimensional problems.

Projectile motion near Earth is the motion of an object after launch when air resistance is neglected. Horizontal acceleration is zero, while vertical acceleration is approximately constant and downward. The horizontal and vertical motions share the same time but otherwise evolve independently. This is why a horizontally launched object and a simultaneously dropped object reach level ground at the same time, provided they begin at the same height.

The trajectory is parabolic only under the ideal assumptions of constant downward acceleration, flat ground, and no drag. Real projectiles deviate because gravity varies slightly, air exerts forces, and Earth rotates, but the ideal model is powerful because it isolates the dominant structure.

$$A_x = A \cos \theta, \quad A_y = A \sin \theta$$

Components for an angle measured from the positive x-axis.

$$x(t) = x_0 + v_0 \cos \theta \cdot t$$

Horizontal projectile motion without drag.

$$y(t) = y_0 + v_0 \sin \theta \cdot t - \frac{1}{2}gt^2$$

Vertical projectile motion with upward chosen positive.

CONCEPT CHECK

- Why is the horizontal velocity constant in the ideal projectile model?
- Which launch angles give the same range on level ground when drag is absent?

WORKED EXAMPLE**Launch from level ground**

A ball is launched at 20 m s^{-1} at 30° above horizontal. Neglect air resistance and take $g = 9.8 \text{ m s}^{-2}$. Find flight time and range.

1. Resolve the initial velocity: $v_{0x} = 20 \cos 30^\circ = 17.3 \text{ m s}^{-1}$; $v_{0y} = 20 \sin 30^\circ = 10.0 \text{ m s}^{-1}$.
2. At landing $y - y_0 = 0$, so $0 = v_{0y}t - \frac{1}{2}gt^2$. The nonzero solution is $t = 2v_{0y}/g = 2.04 \text{ s}$.
3. Range $R = v_{0x}t = 17.3(2.04) = 35.3 \text{ m}$.

Result. Flight time $\approx 2.04 \text{ s}$; range $\approx 35.3 \text{ m}$.

Interpretation. The vertical component determines how long the projectile remains aloft; the horizontal component converts that time into range.

MODULE 01 • CONCEPTUAL DEVELOPMENT

1.4 Newton's laws and free-body diagrams

Newton's first law identifies inertial reference frames: frames in which an object with zero net force maintains constant velocity. The law is not merely the special case $F = 0$ of the second law; it specifies the kind of frame in which the standard form of the second law is valid.

Newton's second law states that the vector sum of external forces on an object equals the time rate of change of its momentum. For constant mass this becomes $\Sigma \mathbf{F} = m\mathbf{a}$. The acceleration points in the direction of the net force, not necessarily in the direction of the velocity. A free-body diagram is the bridge between the physical situation and this equation.

Newton's third law concerns interactions: if object A exerts a force on B, B simultaneously exerts an equal-magnitude, opposite-direction force on A. Third-law partners act on different objects and therefore never cancel on one object's free-body diagram. Weight and normal force are often equal in magnitude, but they are not a third-law pair.

$$\Sigma \mathbf{F} = d\mathbf{p}/dt$$

General form of Newton's second law.

$$\Sigma \mathbf{F} = m\mathbf{a}$$

Constant-mass form.

$$\mathbf{F}_{A \rightarrow B} = -\mathbf{F}_{B \rightarrow A}$$

Newton's third-law interaction pair.

CONCEPT CHECK

- Why must every force on a free-body diagram have an identifiable source?
- Why do third-law forces not cancel in the equation for a single object?

WORKED EXAMPLE

Two blocks pulled together

Blocks of 2.0 kg and 3.0 kg are connected by a light cord on a frictionless table. A 20 N horizontal force pulls the 3.0 kg block. Find the acceleration and cord tension.

1. Treat both blocks as one system: $a = F/(m_1 + m_2) = 20/5.0 = 4.0 \text{ m s}^{-2}$.
2. Isolate the 2.0 kg block. Its only horizontal force is the tension T .
3. Apply $T = m_1 a = 2.0(4.0) = 8.0 \text{ N}$.

Result. Acceleration = 4.0 m s^{-2} ; tension = 8.0 N.

Interpretation. Choosing the entire system first eliminates the internal tension; isolating one block afterward reveals it.

MODULE 01 • CONCEPTUAL DEVELOPMENT

1.5 Common forces and a repeatable solution method

Near Earth, weight is the gravitational force mg . A normal force is a contact force perpendicular to a surface; it is not automatically equal to weight. Tension is the force transmitted by an ideal light cord and acts away from the object along the cord. Spring force is approximately proportional and opposite to displacement from equilibrium. Friction and drag require additional models developed in Module 2.

A reliable Newtonian solution follows a fixed sequence: define the system; choose axes; draw the free-body diagram; translate the diagram into component equations; solve symbolically when possible; then inspect units, signs, limiting cases, and magnitude. Skipping the diagram often produces missing forces or forces assigned to the wrong object.

The final numerical value is not the whole answer. A negative result may indicate a direction opposite to the assumed positive axis rather than an error. An impossible limiting case, wrong unit, or acceleration wildly inconsistent with the available force signals a model or algebra mistake.

$$W = mg$$

Weight near Earth's surface.

$$F_s = -kx$$

Hooke's law for an ideal spring within its elastic range.

CONCEPT CHECK

- When can a normal force be smaller than, equal to, or larger than weight?
- What checks should be performed after solving a force problem?

WORKED EXAMPLE

Apparent weight in an elevator

A 70 kg person stands on a scale in an elevator accelerating upward at 1.5 m s^{-2} . What does the scale read?

1. For the person, choose upward positive. Forces are normal force N upward and weight mg downward.
2. Newton's second law gives $N - mg = ma$.
3. Thus $N = m(g + a) = 70(9.8 + 1.5) = 791 \text{ N}$.

Result. Scale reading = $7.9 \times 10^2 \text{ N}$.

Interpretation. The scale measures the normal force, not gravity directly. Upward acceleration makes the apparent weight exceed mg .

MODULE 01

Guided Exercises

Complete the reconstruction before consulting the separate instructor guide. Diagrams, assumptions, units, and interpretation are part of the solution.

GUIDED EXERCISE

Exercise 1A — Motion reconstruction

A particle's velocity is $+4 \text{ m s}^{-1}$ from 0–3 s, decreases linearly to -2 m s^{-1} from 3–6 s, and remains -2 m s^{-1} from 6–9 s. Draw the velocity-time graph, calculate total displacement, identify turning time, and sketch a compatible position-time graph.

REQUIRED MOVES

- Use signed area under the velocity graph.
- Locate the instant at which velocity crosses zero.
- Make the position graph's slope match the velocity in each interval.

GUIDED EXERCISE

Exercise 1B — Free-body diagram clinic

For each situation—book on a table, sled pulled by an angled rope, falling ball with drag, and two-block pulley system—draw a separate free-body diagram for the named object. Label the source of every force and identify all third-law partners outside the diagram.

REQUIRED MOVES

- Circle the system before drawing forces.
- Include only forces acting on that system.
- Explain why velocity arrows and ‘centripetal force’ do not belong as extra forces.

MODULE 01 - EXPANDED TUTORIAL STUDIO

Module 1 - Interactive Deepening

Newtonian Description of Motion and Force

Purpose. This studio extends the core exposition with derivation practice, worked examples, tutorials, an investigation, computational modelling, error diagnosis, an unguided problem set, and a mastery test. Attempt every prompt before consulting the solutions volume.

From descriptions to dynamical models

Introductory mechanics becomes reliable when every problem is treated as a chain of modelling decisions rather than a hunt for a matching formula. First define the system, time interval, reference frame, and approximation. Then choose coordinates and representations. Only after this should equations be written. The same physical event can be represented by a position graph, a velocity graph, a vector diagram, a free-body diagram, or a differential equation; each representation exposes different information.

Kinematics is most general when written with derivatives and integrals. Velocity is $v_x = dx/dt$ and acceleration is $a_x = dv_x/dt = d^2x/dt^2$. Constant-acceleration equations are therefore special integrated cases, not independent laws. For non-constant acceleration the graph, integral, or differential equation is the correct tool. This distinction prevents one of the most common errors in first-semester physics: using a constant-acceleration identity when the force depends on position or velocity.

Newtonian dynamics adds an interaction model. The statement $\sum \vec{F}_{\text{ext}} = d\vec{p}/dt$ becomes $\sum \vec{F} = m\vec{a}$ only for constant mass in an inertial frame. A free-body diagram is a model of interactions acting on the selected system, not a picture of everything present in the scene. The diagram must contain forces, not velocities, accelerations, energy arrows, or invented labels such as centripetal force unless a real interaction supplies that inward force.

Circular motion is a particularly useful test of understanding. Constant speed does not imply zero acceleration because velocity is a vector. For uniform circular motion the acceleration magnitude is $a_c = v^2/r = \omega^2 r$ and points toward the centre. The inward net force may be tension, gravity, friction, a normal force, or a combination. Treating centripetal force as an extra force double-counts the dynamics.

Derivation board

Kinematics from calculus

$$v_x = \frac{dx}{dt}, \quad a_x = \frac{dv_x}{dt}, \quad \Delta x = \int_{t_i}^{t_f} v_x dt, \quad \Delta v = \int_{t_i}^{t_f} a_x dt$$

Uniform circular motion

$$\left| \frac{d\vec{v}}{dt} \right| = \frac{v^2}{r} = \omega^2 r$$

Worked-example studio

Non-uniform acceleration from a velocity law

A cart has velocity $v(t) = 2.0 + 0.80t^2$ m/s. Find acceleration at 3.0 s and displacement from 0 to 3.0 s.

Solution path

1. Differentiate: $a(t) = 1.60t$, so $a(3) = 4.8$ m/s².
2. Integrate velocity: $\Delta x = \int_0^3 (2 + 0.8t^2) dt = 2t + (0.8/3)t^3 \Big|_0^3$.
3. The displacement is 13.2 m.

Interpretation. The calculation uses the definitions directly; no constant-acceleration formula is valid because acceleration changes with time.

Horizontal curve and friction

A 900 kg car moves at 15 m/s around a flat curve of radius 50 m. What minimum static-friction coefficient prevents slipping?

Solution path

1. Required inward force is mv^2/r .
2. The maximum static friction is $\mu_s mg$.
3. Set $\mu_s mg \geq mv^2/r$, giving $\mu_s \geq v^2/(rg) = 0.459$.

Interpretation. Mass cancels. The friction is not an additional centripetal force; it is the interaction that provides the inward net force.

Tutorial workshops

Tutorial A - Graph reconstruction clinic

Work in the order shown. Do not skip the prediction or representation stages.

- 1. Sketch an x - t graph with three intervals: increasing concave up, decreasing linear, and horizontal.**
- 2. From your sketch, produce a qualitative v - t graph and then an a - t graph.**
- 3. Mark every instant at which velocity is zero and every interval of positive acceleration.**
- 4. Explain which features are invariant if the origin of position is shifted.**

Tutorial B - Constraint and force model

Work in the order shown. Do not skip the prediction or representation stages.

- 1. A block on a horizontal table is connected over a pulley to a hanging block. Draw a separate free-body diagram for each block.**
- 2. Choose one coordinate direction that makes the acceleration constraint explicit.**
- 3. Derive the acceleration and tension symbolically for masses m_1 and m_2 with a frictionless pulley.**
- 4. Then add kinetic friction on the table and identify exactly where the derivation changes.**

Debug the solution. A student writes: "The car moves at constant speed, so acceleration is zero. Therefore the inward friction force on a circular road must also be zero." Identify both conceptual errors and rewrite the force equation correctly.

Required response: identify the first invalid step, state the violated principle or assumption, and repair the solution from that point onward.

Experimental and computational investigation

Investigation - Video motion tracking

Record a safe one-dimensional motion with a phone at known frame rate. Calibrate distance with a ruler or known object, extract at least 20 position-time points, estimate velocity by central differences, estimate acceleration, and compare the numerical derivative with a model. Report scale and timing uncertainty and explain how differentiation amplifies noise.

Minimum evidence: raw observations or simulation settings; model equation; graph; uncertainty or numerical-error statement; comparison between prediction and result; brief limitations paragraph.

Computational mini-lab - Numerical kinematics from measured acceleration

Starting from a sampled acceleration series, use a time step Δt to update velocity and position by $v_{n+1} = v_n + a_n \Delta t$ and $x_{n+1} = x_n + v_n \Delta t$. Compare the numerical trajectory with an analytic constant-acceleration case and quantify the time-step error.

Reproducibility rule: state initial conditions, parameter values, units, algorithm or code, step size where relevant, and a numerical or limiting-case validation.

Problem set

These problems are deliberately less scaffolded than the guided exercises. Draw a representation before calculation; solve symbolically when practical; carry units; and perform at least one check.

1. A particle moves as $x(t) = 4 - 3t + 2t^2$ m. Find velocity and acceleration at 2 s.
2. A runner travels 300 m east in 60 s and 100 m west in 20 s. Find average velocity and average speed.
3. A projectile leaves level ground at 22 m/s and 35 degrees. Neglect drag. Find time of flight, maximum height, and range.
4. A 12 kg crate is pulled by a 70 N rope 25 degrees above horizontal on a frictionless floor. Find horizontal acceleration and normal force.
5. Two blocks 4 kg and 6 kg are connected on a frictionless table and pulled by 50 N. Find acceleration and tension when the force acts on the 6 kg block.
6. A 0.25 kg ball on a 0.80 m string moves at 5.0 m/s in a horizontal circle. Find the required inward force.
7. Derive the constant-acceleration relation $v_f^2 = v_i^2 + 2a\Delta x$ by eliminating time.
8. Explain with a diagram why normal force and weight are not a third-law pair.
9. A scale in an elevator reads 620 N for a 60 kg person. Find the elevator acceleration and direction.
10. Construct a dimensional check for $R = v_0^2 \sin(2\theta)/g$.

Module mastery test

Closed notes. Suggested time: 25 minutes. Scientific calculator permitted. A correct number without a visible model, units, or explanation may receive only partial credit.

1. The slope of a position-time graph is

- A. acceleration**
- B. velocity**
- C. displacement**
- D. force**

2. The area under an acceleration-time graph gives

- A. position**
- B. change in velocity**
- C. distance**
- D. force**

3. In uniform circular motion, acceleration points

- A. tangentially**
- B. outward**
- C. toward the centre**
- D. opposite velocity**

4. A free-body diagram contains

- A. only external forces on the system**
- B. all forces the system exerts**
- C. energy arrows**
- D. all nearby objects**

5. Newton third-law forces

- A. act on one object and cancel**
- B. act on different objects**
- C. are always weight and normal**
- D. require contact**

6. If velocity is negative and acceleration positive, the object is

- A. always speeding up**
- B. always slowing down at that instant**
- C. stationary**
- D. moving in a circle**

7. State the modelling steps that should occur before substituting numbers.

8. Explain why constant speed can coexist with nonzero acceleration.

MODULE 01

Module Quiz

Answer without notes. For constructed responses, a correct equation without explanation is incomplete. Solutions are in the separate instructor guide.

1

A runner completes one lap and stops at the starting point. Which quantity must be zero?

- | | |
|-----------------|------------------|
| 1. distance | 2. average speed |
| 3. displacement | 4. elapsed time |

2

An object moves in the +x direction while its acceleration is negative. It must be...

- | | |
|----------------|---------------------------------|
| 1. speeding up | 2. slowing down at that instant |
| 3. at rest | 4. moving in a circle |

3

The area under a velocity-time graph gives...

- | | |
|-----------------|--------------------|
| 1. acceleration | 2. distance always |
| 3. displacement | 4. jerk |

4

In ideal projectile motion, which quantity is constant?

- | | |
|------------------------|--------------------------|
| 1. vertical velocity | 2. speed |
| 3. horizontal velocity | 4. direction of velocity |

5

A correct free-body diagram for one object includes...

- | | |
|---------------------------------|--|
| 1. all forces the object exerts | 2. all external forces on the object |
| 3. its velocity as a force | 4. third-law pairs on the same diagram |

6

The normal force on a box is always equal to mg .

- | | |
|---------|----------|
| 1. True | 2. False |
|---------|----------|

7

State Newton's third law without using the vague phrase 'every action has a reaction.'

8

Why is $\Sigma F = ma$ insufficient until a system and frame have been chosen?
